Check for updates

INTERNATIONAL JOURNAL OF CLIMATOLOGY
Int. J. Climatol. **35**: 3486–3504 (2015)
Published online 5 December 2014 in Wiley Online Library
(wileyonlinelibrary.com) DOI: 10.1002/joc.4223

RMetS
Royal Meteorological Society

The Oslo temperature series 1837–2012: homogeneity testing and temperature analysis

Ø. Nordli,[a]* G. Hestmark,[b] R. E. Benestad[a] and K. Isaksen[a]

[a] *Research and Development Department, Division for Model and Climate Analysis, Norwegian Meteorological Institute, Blindern, Oslo, Norway*
[b] *Department of Biosciences, Centre for Ecological and Evolutionary Synthesis, University of Oslo, Blindern, Norway*

ABSTRACT: A 175 years long homogenized composite record of monthly mean temperatures is presented for Oslo, the capital of Norway. The early raw data have been digitised and quality controlled, and monthly means have been calculated. Some early original observations carried out in a Wild screen (1877–1936) were found to be spuriously high because of inappropriate sheltering from sunlight. These spurious temperatures were not used in the composite record, but alternative temperatures measured (1837–1933) by thermometers placed outside windows at the Astronomical Observatory were used instead. No inhomogeneity was detected in the latter series after adding an instrument correction of +0.3 °C, but the start year of the correction remains uncertain. The more recent part of the composite record used the long-term series (1937 to present) from Blindern in Oslo, the premises of The Norwegian Meteorological Institute. Two small inhomogeneities were detected in the Blindern series, possibly caused by a weak urban heat island effect or growing/cutting of trees. The study revealed that the annual mean temperature has increased by 1.5 °C in the period 1838–2012. The most pronounced increase in annual temperature occurred during the last 50 years, and in the early 20th century that ended with a local maximum in the 1930s. The temperature has increased significantly in all seasons; however, the temperature increase in summer was less than a half of that in winter and spring, which were the seasons with largest increase. In addition the monthly mean temperature of the coldest month in each year has increased two times faster than the warmest one. The most significant temperature variations were associated to ∼5-year time scales in its early part, but since 1930 and up to present, the dominant time scales were 10–20 years.

KEY WORDS Long-term temperature series; Oslo University Astronomical Observatory; Station history; Köppen's formula; Føyn's formula; The early instrumental warm bias

Received 5 March 2014; Revised 31 October 2014; Accepted 31 October 2014

1. Introduction

How has the air temperature of the Norwegian capital varied over the past two centuries? This is the main question we want to address by the present article, but in order to obtain a reliable answer, a homogenous time series is needed. Hence our focus will be on homogeneity testing. We present a homogenized series based on a composite of three long-term series from Oslo. There has been a growing interest in long-term air temperature series with the current concern and debate about climate change and global warming. Our goal is local/regional but nevertheless each long-term series provides important pieces of information about anthropogenic global warming and natural climate variability. For examples the datasets used for the calculation of the global mean land–surface air temperature are steadily increasing by including additional series over time (Hartmann *et al.*, 2013, p. 187). More long term series improve the reliability of global temperature and thereby the trends in those datasets.

Homogenizing long-term climate time series is a challenging undertaking as all available series are in general inhomogenous. The reason for this may be changes in instruments (precision, mounting, exposure), observation procedures (observation times), station surroundings or relocations, subjective factors (individual observer bias). If not adjusted for, such inhomogeneities may affect estimates of trends of both mean climate and climate extremes, and lead to unwarranted conclusions (Moberg and Alexandersson, 1997; Tuomenvirta, 2001; Aguilar *et al.*, 2003, pp. 38–43; Auer *et al.*, 2007). When the national meteorological institutes were established in the last part of the 19th century, fixed national standards were introduced both for observational procedure, observation times and for screening the thermometers to avoid exposure to sunlight, that is, radiation screens, worldwide (Parker, 1994) as well as in the Nordic countries (Nordli *et al.*, 1997). The national institutes built up station networks, which have been of fundamental help for homogeneity testing as now neighbouring stations were available for comparison, that is, one series could be tested relative to its neighbours. The standardized networks of stations, better instrument

* Correspondence to: Ø. Nordli, Research and Development Department, Division for Model and Climate Analysis, Norwegian Meteorological Institute, P.O. Box 43, Blindern, 0313 Oslo, Norway. E-mail: oyvind.nordli@met.no

© 2014 Royal Meteorological Society

and observational techniques were achievements that led to the end of what is called the early instrumental epoch in the last part of the 19th century (Camuffo, 2002a, 2002b; Böhm *et al.*, 2010, p. 42).

For central Europe a regional homogenization of temperature series from the Alps based on no less than 97 stations was performed by Böhm *et al.* (2001). In the Netherlands, the De Bilt series, starting already in 1735, is still in operation, but other even older series exist in the Netherlands that have been tested and corrected (van Engelen and Geurts, 1985), and early observations from central Belgium, starting 1767, have also been homogenized (Demarée *et al.*, 2002). In Germany, the Hohenpeissenberg series, which is of particular interest as it is in the countryside free from urbane heat island influence, goes back to 1781 (Winkler, 2009). However, the longest in the world is the Central England Temperature (CET) series starting in the 1659 (Manley, 1953, 1974; Parker and Horton, 2005), now continually updated by the Met Office Hadley Centre. For the Nordic countries we possess several Swedish series, in particular the two famous neighbouring temperature records from Uppsala (beginning in 1722) and from Stockholm (beginning in 1756) that have been homogenized (Moberg and Bergström, 1997; Bergström and Moberg, 2002; Moberg *et al.*, 2002). Relevant to this latitude is also the St Petersburg series in Russia starting in 1743 (Jones and Lister, 2002). For a long-term monthly mean data set (the so called NACD-series starting in 1890) from the Nordic countries, the inhomogeneities were found to be less than 0.5 °C with very few exceptions (Tuomenvirta, 2001). Also temperature series averaged over grid boxes covering mainly Sweden have been homogenized (Moberg and Alexandersson, 1997).

Pioneering work was done by the climatologist Birkeland (1926, 1930, 1931, 1935, 1949) on the early Norwegian instrumental series who did quality control, and also published homogenized series of monthly mean values for pressure and temperature. For many of the series he encountered great problems due to lacking knowledge of observation times, radiation screening and instrument corrections. The least reliable parts of the series were, as expected, those extending from the start until their qualities were improved in the middle of the 19th century. In particular those periods are: for Trondheim, 1762–1855 (Birkeland, 1949), Bergen 1818–1858 (Birkeland, 1930), Ullensvang 1799–1871 (Birkeland, 1931). Nordli (2001) who compared Birkeland's homogenized Trondheim series with reconstructions from proxy data, concluded that it still contained inhomogeneities that were connected to relocations and changes of observers. These deficiencies in the early Norwegian series may be explained by the lack of permanent institutions, like monasteries and old universities. Without those institutions, the country's oldest data were naturally the results of private initiatives. But for Oslo, which is the subject for this article, the situation is different as observations at fixed hours started at the Astronomical Observatory in 1837 (Birkeland, 1926).

The epoch we have called early instrumental observations came to an end in Norway with the founding of the Norwegian Meteorological Institute (MET Norway) 1 December 1866. A standardized nationwide station network was soon established, and already in 1875, 38 stations were in operation. Homogeneity testing of long-term series for the period 1876–1996 was performed by Nordli (1997), who found that out of 22 series, 13 were considered to have inhomogeneities successfully corrected for. The most likely reasons for inhomogeneities were associated with relocations (37%) and sunshine on the wall thermometer cage (24%). Homogeneity testing has also recently been performed on a series from the Agricultural College (now university) at Ås, 30 km to the south-east of Oslo, where five inhomogeneities were detected (Gjelten, 2011). An overall homogeneity analysis for the whole of Norway has recently been undertaken by Andresen (2011).

Our objective was to study long-term variations of homogenized monthly mean temperatures in Oslo. The project HistKlim (History of Climate) at MET Norway has made it possible to digitize raw data from Oslo, perform extensive raw data quality control, and calculate all monthly mean values. We consider monthly values to be an adequate temporal resolution. Here we first present the station history and metadata for the Oslo series, followed by a presentation of the methods used. We then continue with the results of the homogeneity testing, and finally an analysis of the long-term temperature evolution.

2. Station history

The Oslo temperature series presented here is a composite based on three series of observations, together spanning a time interval from 1837 to 2012. Two of the series – *18650 Oslo I* and *18651 Oslo II* referred to by their local station identifiers and names – are from the site of the old Astronomical Observatory of the University of Oslo. The station name and their roman numbering reflect the fact that MET Norway numbered their Oslo station with the numeral *I*, despite an earlier start of the observation from the Astronomical Observatory; thus *Oslo II* starts before *Oslo I*. They substantially overlap in time (Figure 1, Table 1) but were made with different instruments. The third series of observations was made at the current site of MET Norway at Blindern, Oslo, station identifier *18700* (Figure 1, Table 1).

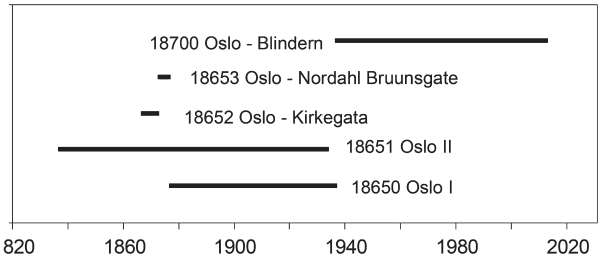


Figure 1. The Oslo stations and their periods of observation. Their geographical locations are shown in Figure 4.

© 2014 Royal Meteorological Society

10970088, 2015, 12, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4223 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 1. Stations in operation near the city centre of Oslo during the period from 1837 to present: national station number (identifier) and name, period of observation, station altitude (H_s), observation times (UTC) in given periods.

No. and name	Period (from–to; year, month, day)	H_s(m)	Obs. times (UTC) period (from–to; year, month, day)
18650 Oslo I	1877.01.01–1937.04.30	25	07,13,19: 1877.01–1920.06[a] 07,13,18: 1920.07–1937.04
18651 Oslo II	1837.04.02–1933.12.31	25	08,13,15,21:1837.04–1838.02[a] 06,08,13,15,21:1838.03–1927.06[a] 07,13,18: 1927.07–1933.12
18652 Oslo–Kirkegata	1866.12.01–1872.10.18	23	07,13,19: 1866.12–1872.10[a]
18653 Oslo–Nordahl Bruunsgate	1872.10.19–1876.12.31	41	07,13,19: 1872.10–1876.12[a]
18700 Oslo–Blindern	1937.03.01 to present	94	07,13,18: 1937.03–1948.12 07,12,18: 1949.01–1960.12 06,12,18: 1961.01 to present

[a]The exact observation times may be UTC time minus 17 min.



Figure 2. The Astronomical Observatory was the site of the temperature series 18651 Oslo II. The photo (from about 1869, Oslo City Museum, http://www.oslobilder.no/OMU/OB.OT056) shows the main entrance at the northern wall. An unsheltered thermometer is visible at the ground floor to the left of the entrance. This window is in room No. 4 at Figure 3.

Oslo II – 18651. The oldest observations we used go back to 1837, and were made by the observers at the Astronomical Observatory of the University of Christiania (now Oslo) (Figures 2 and 3). The Observatory was completed in 1833, and located at the site Solli, in the countryside well outside town. Since then Oslo has expanded far beyond the site of the Observatory, which is today regarded as being located close to the city centre (Figure 4). The building has recently been converted to a museum for the history of science. In a description of the Observatory by its first Director Prof. Christopher Hansteen and its second Director Carl Fearnley (Hansteen and Fearnley, 1849, p. 2, 5), it is stated that the barometer was located in a north-facing room at the ground floor, and a thermometer by the Berlin instrument maker Carl P. H. Pistor outside the only window of the room (Figure 3). Two other thermometers, by the Berlin instrument maker J. G. Greiner Jr., were located outside the offices of the Director and the observer, that is, rooms No. 6 at the eastern wall and No. 8 at the western wall. The use of these three thermometers is confirmed by Fearnley (1865, p. VI). All thermometers were scaled in degree Reaumur. Continuous meteorological observations at the Observatory started on 2 April 1837, and were until 28 February 1838 only four times a day: 0900, 1400, 1600 and 2200. From 1 March 1838 one observation at 0700 was added. All observations were noted in so-called Christiania Time. A national reference meridian was defined with the establishment of the Observatory based on the longitude of the building: 10°43′E relative to Greenwich. Christiania time was introduced according to this meridian, being 43 min earlier than Greenwich Mean Time (UTC). From 1 January 1895 Central European Time (CET) was introduced in Norway, but it is not known whether the observation times at the Observatory were changed accordingly. For convenience the observation times are referred to in UTC (e.g. 1300 UTC) in this article neglecting the possible difference of 17 min. The first Director of MET Norway (Mohn, 1895, p. 5) wrote that the Observatory series is entirely homogeneous as the thermometers have remained in the same places, and the environment has not changed. He confirms the information in Fearnley (1865) and Hansteen and Fearnley (1849) concerning the sites of the instruments. According to Mohn, the thermometers were calibrated by Hansteen, but 'they were not sheltered from precipitation' (Mohn, 1895, p. 5). The unsheltered thermometer is seen on a photo from around 1869 of the north wall (Figure 2), also showing the position of the bulb of the thermometer which by our measurement at the site in 2012 was estimated to be 2.8 ± 0.1 m above the ground. The eastern and western windows were lower above the ground so that the heights of the two other thermometers were only 1.8 ± 0.1 m if their bulbs were placed like that at the north wall. Birkeland (1926, p. 18) stated that in the beginning observations were made on all three thermometers, on the W, N and E walls, but infrequently on the latter. The thermometers were calibrated by Hansteen using the 'method of Bessel' (Fearnley, 1865, p. VI; Birkeland, 1926, p. 18). In Hansteen's 'correction book' (book of calibrations), kept at the National Archives in Oslo (S-1570/F/Fa), only two thermometers are recorded, Greiner I and Greiner II. In the collection of old instruments at the Department of

© 2014 Royal Meteorological Society

10970088, 2015, 12, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4223 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

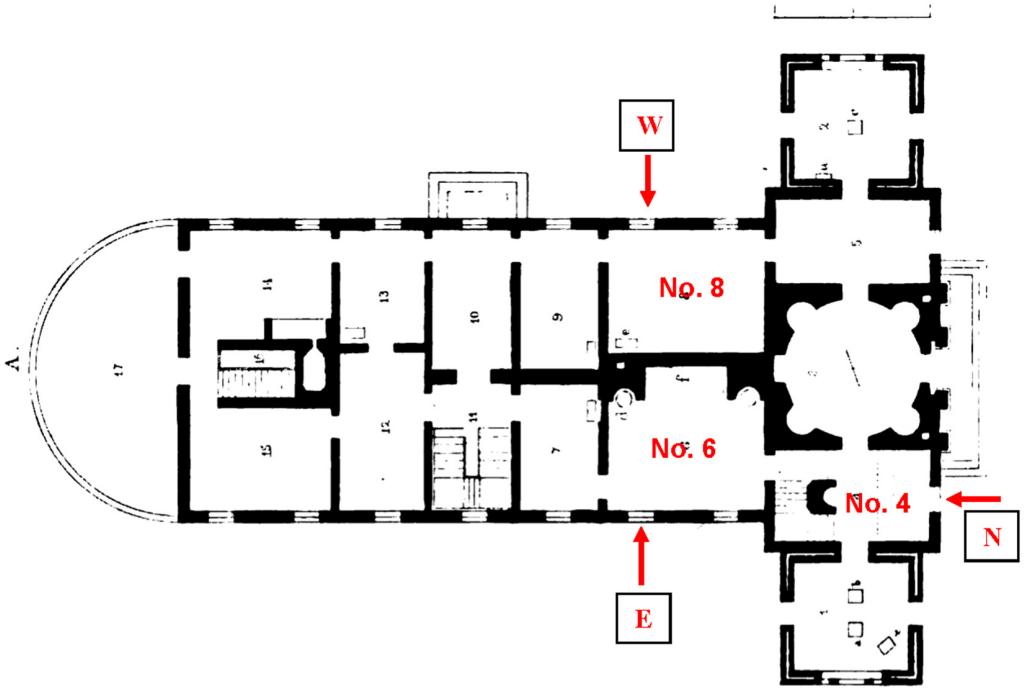


Figure 3. The ground floor of the Observatory building, the site of 18651 Oslo II (Hansteen and Fearnley, 1849) with some remarks added (bold). In 1849 and 1887, a thermometer of Pistor type was placed outside the window of room No. 4 at the northern wall (N), whereas thermometers of Greiner type were placed outside eastern (E) and western (W) windows in room Nos. 6 and 8.

Physics at the University of Oslo two old Greiner thermometers are preserved and these are likely to be the thermometers used at the Observatory. Increasing air pollution from the encroaching town made the Observatory close on 31 December 1933, ending the nearly 100 year long meteorological observation series. Digitalization of the Oslo II series is based on Fearnley's printed record for the period 1837–1867 (Fearnley, 1867) and from handwritten observation protocols 1868–1933, preserved at MET Norway.

Oslo I – 18650. In 1877 MET Norway set up its instruments on a hill adjacent to the eastern wall of the Observatory building, commencing a parallel observation series that would run to 1937 (Hesselberg, 1917, p. 31). The thermometers were placed in a 'Wild screen', named after Heinrich Wild, Director of the Central Physical Observatory in St Petersburg. The height above the ground for the thermometer was 2 m (Mohn, 1895, p. 6), in accordance with the minimum height recommended by Wild, whereas Wild himself designed screens that were even higher (Middleton, 1966, pp. 227–230). The Wild screen series continued until April 1937, and thus overlap substantially in time the observations made at the adjacent Observatory until its closure in 1933 (cf. Table 1). From 1 July 1920 (Table 1) the evening observations were made 1 h earlier than before, which have to be compensated for when monthly mean temperature is calculated. For stations belonging to MET Norway, like *Oslo I*, monthly means are routinely calculated by the so called Köppen's formula. The formula is described in some details in Appendix A1.

Previous to *Oslo I* MET Norway ran a couple of brief observation series at its first offices in the centre of town: *18653 Oslo–Kirkegata* and *18654 Oslo–Nordahl Bruunsgate*, station names denote the streets where observations were taken (Figure 1, Table 1). At Kirkegata the instruments were placed on a northern wall outside the third floor (Mohn, 1867, p. IV). In October 1872 the institute moved to Nordahl Bruunsgate 8, and the instruments were placed in a tower room on the roof at a more free position (Mohn, 1872, Introduction). The thermometers were screened by wall or window cages (Mohn, 1876, p. III). However, rapid expansion of the city made these observations more and more influenced by buildings, spill heat, and pollution, and their questionable quality and comparatively brief time spans made us exclude them in the present study.

18700 Oslo–Blindern: On 1 March 1937 a new observation series was started at the site where the present premises of MET Norway was planned (Figure 4), construction work having started in 1937/1938 and near finished in April 1940 (Barlaup, 1966, pp. 65–66). The Blindern observations were taken on a small hill (Figure 5), about 75 m to the southwest of the institute building, on the same spot as the observations are still carried out. On 8 October 1951 the temperature screen was changed from the pattern of 1930 (MI-30) to the pattern of 1946 (MI-46) (Nordli *et al.*, 1997). Other changes that could influence temperature measurements were not noted in the metadata file, but we know that trees in the sector south to west have continually been growing (Figure 5) and that some of them occasionally have been cut down without knowing exactly when. The observation time has been changed twice (Table 1), but this is compensated for by calculation of monthly mean temperature by Köppen's formula (Appendix A1), see also comment under *Oslo I*.

© 2014 Royal Meteorological Society

10970088, 2015, 12, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4223 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License



Figure 4. Map of Oslo showing the four measuring sites: (a) Kirkegata, (b) Nordahl Bruunsgate, (c) Astronomical Observatory, the site of both Oslo I and Oslo II, and (d) Blindern. The map shows the situation at the end of the 1950s: the Astronomical Observatory was completely surrounded by buildings, whereas most of the projected university buildings at Blindern still were not built.

3. Methods

3.1. The estimation of monthly mean temperature

Our main goal was to develop a homogeneous temperature series for Oslo with monthly values, so it was important that monthly mean estimates were calculated as accurate as possible and with minimum biases when only a few observational times exist during the day (Tuomenvirta, 2001). In Norway the so called Hansteen's, Føyn's and Köppen's formulas have been used. They are based on three different principles: Hansteen's formula (Appendix A1) is used when there are more than three observations unevenly distributed during the day. It weights the observations according to the time interval between them. Føyn's formula (Appendix A1) is used for three observations during the day, morning, midday and evening. It averages the morning and evening observations, which do not differ much from the monthly mean value, whereas the midday observation is used only as an adjustment of the results, not being heavily weighted. Köppen's formula (Appendix A1) is for three observations (nearly) evenly distributed during the daytime equally weighted, whilst the daily minimum temperature compensates for lacking night observations.

All formulas for calculation of monthly means include coefficients, which have to be determined from hourly observations. This is done by MET Norway for their own stations so we have adopted the monthly means from their database (http://www.eklima.no; last accessed 31 October 2014). For 18651 Oslo II, belonging to Astronomical Observatory, this has not been done since Birkeland (1926), who had hourly observations from the Observatory to his disposal spanning over an interval of only 32 months (Old Dataset: 1841.11–1843.06 and 1846.08.15–1847.08.14). Now a much larger hourly dataset from Blindern (New Dataset: 1992.12–2009.06) is available so the calculation of the coefficients is revisited.

© 2014 Royal Meteorological Society

10970088, 2015, 12, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4223 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License



Figure 5. The environment of the station 18700 Oslo–Blindern with its MI-46 screen in the centre of the photo. To the south and west (seen in the photo) the screen is surrounded by trees, to the east and north the terrain is more open (photo Ø. Nordli).

Birkeland (1926) used Føyn's formula (Nordli and Tveito, 2008), (Appendix A1) although the series has four or five daily observations, which he formally reduced to three by taking the mean of observation times 0600 and 0800 UTC as the morning observation, 1300 and 1500 UTC as the midday observation, whereas 2100 UTC remained the evening observation. Both Hansteen's and Føyn's formulas will be tested against the New Dataset of hourly observations.

For control of the results the hourly dataset are divided into two separate datasets, one for training and one for validation. The coefficients were estimated from the training data, and the validation data were used to test the monthly mean values. The accuracy of monthly mean estimates based on only a few daily observations is evaluated by taking the difference to the 'true means' based on all 24 hourly observations. However, both the training as well as the validation dataset was used for the calculation of the final coefficients. The coefficients in the formulas need to be changed accordantly if the observational procedure is changed, for example, observation times. Accurate coefficients are of crucial importance since incorrect values will lead to inhomogeneities.

3.2. Methods and procedures for homogeneity testing

The procedure for homogeneity testing followed the principle of comparing a candidate series (the series under testing) against reference series. Traditionally the reference might be series from one or more neighbouring stations. A candidate series might also be observations at one particular time of the day, which are compared with observations at other times of the day for the same station. This procedure we have called 'internal testing'. For a long period of time, 1837–1866, there was only one station in operation in south-eastern Norway (Figure 1), so internal testing was the only possibility. The homogeneity testing methods (Appendix A3) chosen for the internal testing was the *MLR* (Vincent, 1998) and *SNHT* (Alexandersson, 1986). In our particular case, the candidate series consisted of values from one observational time, which was compared with the mean values of the other observational times, and the testing was performed with seasonal resolution, December–February (DJF), March–May (MAM), June–August (JJA) and September–November (SON). When appropriate, additional information was sought also by using monthly resolution, and by making different selections for the references rather than the mean values of all of them.

In the 1870s a network of stations was established in south-eastern Norway. This made it possible to use the *HomeR* software package for homogeneity testing and 'correction' of the series. The software is a result of the EU-funded project COST Action ES0601. (In later years, and in particular among researchers working in the COST Action the terminology used is 'correction' rather than 'adjustment'. One has, however, to keep in mind that corrections in this context not necessarily mean correcting wrong values, but adjusting the values in order to obtain homogeneity). A practical guide to the program is found in the study of Mestre and Anguilar (2011), whilst the theoretical background is found in the study of Mestre *et al.* (2013) and the references therein. It is assumed that each observation series is a sum of a climate signal, a station effect and random white noise. The station effect is constant if the series is homogeneous, otherwise the series is piecewise homogeneous between the shifts (Caussinus and Mestre, 2004; Mestre *et al.*, 2013). *HomeR*

© 2014 Royal Meteorological Society

10970088, 2015, 12, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4223 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

obtained one of the best results when blind tested against benchmark datasets with known inhomogeneities (Venema *et al.*, 2011). See Appendix A3 for more detailed description about a set of methods used in the homogeneity testing.

3.3. Tools for temperature analysis

Linear trends of temperature time series will be calculated by using time as predictor and temperature as predictand in Least-Squares Regression (Wilks, 1995: pp. 160–176). The homogenized Oslo series will be smoothed by a Gaussian filter looking for non-linear long-term variations in the series, see Appendix A4.

3.4. Time scale analysis

Wavelet analysis was used in the analysis of the temperature series to identify changes in its character over time. There is no *a priory* reason to expect that the dominant temporal scales should be invariant, and the outcome of the wavelet analysis may reflect both natural climate variations, anthropogenic climate change and inhomogeneities. The series was examined for dominant frequencies of variation by wavelet analysis, carried out in R (R Development Core Team, 2004), using the R-packlage 'dplR' with a continuous Morlet wavelet transform (Torrence and Compo, 1998). The wavelet analysis was applied to monthly mean temperature anomalies rather than the original data in order to enhance the inter-annual character. The significance level was set to 0.99 and the number of sub-octaves per octave was set to 0.1.

4. Results

4.1. Calculation of the monthly mean temperature for 18651 Oslo II

There are two alternative formulas that may be used for monthly mean calculation, Hansteen's formula and Føyn's formula (Appendix A1). Using the arithmetic means of the 24 daily observations for the New Dataset (see Section 3.1) as a 'true mean value', the RMSE (Root Mean Square Error) of the anomalies were 0.076 °C for Hansteen's formula and 0.095 °C for Føyn's formula. Further testing of the formulas was done by grouping the Oslo–Blindern dataset in a training dataset and a validation dataset of approximately equal size. Both formulas performed well as no biases were detected when the monthly means were compared with the 'true mean values'.

The Hansteen formula was selected for use in the 18651 Oslo II series because it gave the lowest RMSE. As seen in the Table 2, the constants based on the New Dataset differ only a few hundreds of a degree centigrade from those based on the Old Dataset, with exceptions for August and September.

4.2. Adding instrumental corrections to the Greiner thermometers at 18651 Oslo II

According to metadata the two Greiner thermometers, which had been used at the station since the start, were taken down for calibration in February 1921. Unfortunately both of them had a droplet of mercury at the end of the tube. Due to the droplets the thermometers showed $0.24^{o}R \approx 0.3\,^{\circ}C$ too low (Birkeland, 1926, p. 18). To be sure that these corrections have not been added to the observation protocols for the years before 1921, from which data were digitized, a test by SNHT was readily performed relative to a reference group of the stations Torungen, Ås and Hamar for the period 1883–1933. The test clearly showed that corrections had not been applied. Therefore the corrections were added to the data in the period 1837–1920, although it cannot be stated for sure that the droplets were present already from the start of the series.

Table 2. Constants C_a and C_b in Hansteen's formula (Equations (A6a) and (A6b) in Appendix A1) used for calculation of monthly mean temperature for 18651 Oslo II. The constants are calculated by using an old hourly datasets (1841.11–1843.06 and 1846.08.15–1847.08.14) and by a new one (1992.12–2009.06). The values adopted in this study are from both datasets weighted according to their lengths.

Month	Constant C_a (°C) by Old dataset	Constant C_a (°C) by New dataset	Adopted C_a (°C) 1838.04–1927.06	Adopted C_b (°C) 1837.04–1838.03
1	−0.06	−0.01	−0.02	−0.10
2	−0.08	−0.04	−0.05	−0.18
3	−0.05	−0.13	−0.12	−0.46
4	−0.22	−0.29	−0.28	−0.65
5	−0.40	−0.46	−0.45	−0.85
6	−0.46	−0.45	−0.45	−0.86
7	−0.40	−0.38	−0.38	−0.83
8	−0.25	−0.45	−0.42	−0.68
9	−0.10	−0.21	−0.19	−0.52
10	−0.04	−0.08	−0.07	−0.27
11	−0.02	−0.01	−0.01	−0.06
12	−0.02	−0.14	−0.12	−0.03

4.3. Internal testing of 18651 Oslo II, 1838–1927

As the Oslo II station was the only one near Oslo in the period 1837–1866, internal testing was the only realistic possibility for detection of possible inhomogeneities. Nevertheless the series was tested with this method for a much longer period, 1838.03–1927.06, in which all five observation times were present.

Some potential inhomogeneities were discovered without being consistent, so they were not considered reliable. But one inhomogeneity was detected both with MLR and SNHT in the 2100 UTC observation for spring and summer with break year 1894 (for SNHT summer 1895). The 2100 UTC observation was spuriously warm (0.6 °C for both spring and summer) relative to the other observations from the spring 1895 until the 2100 UTC observations were abandoned 1 July 1927. There is no metadata that could support the detection. Photos from the 1890s are available taken from distance, but as far as it can be seen, no change has taken place, at least not at the northern front of the building, which is most visible at the photos.

© 2014 Royal Meteorological Society

10970088, 2015, 12, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4223 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

The corrected 2100 UTC temperatures led to corrected monthly mean temperatures. For individual months and break year 1894 also for summer, the corrections were estimated to −0.2 °C in May and June, and −0.1 °C in April, July and August, see also formula (A7) (Appendix A2).

4.4. Homogeneity testing of 18651 Oslo II using HomeR software 1877–1933

The testing was performed with two versions of the 18651 Oslo II series, with and without the correction of the inhomogeneity in 1894 discovered by the internal testing (see Section 4.3). The tests showed intriguing results: adding of the corrections from the internal testing caused the Oslo II series to be inhomogeneous, whereas the series turned out to be homogeneous without these corrections. Thus, the result of the internal testing apparently contradicts the result of the testing with reference stations. We put forward two different hypotheses for the possible inhomogeneity of 1894:

- *Hypothesis I*: A discontinuation of the reading from the eastern window thermometer in 1895 affected the calculated averages.
- *Hypothesis II*: A change from Christiania Time to Central European Time (CET) in 1895 affected the calculated averages.

Hypothesis I. There is some support for this hypothesis by Birkeland (1926, p. 18), who related that *the thermometer at the eastern wall is seldom in use*. When he submitted his work (1923) the Observatory was still in operation, so he must be considered as a reliable source. He mentioned that two thermometers were still in operation at the Observatory, both of Greiner type, not three as should be expected, cf. Figure 3. The position of them was in 1923 at the western and northern windows (Figure 3). Hansteen and Fearnley (1849) were specific about the thermometer at the northern window stating that it was of Pistor type, which implied that the thermometer at that window must have been changed. Fearnley also mentioned a Pistor thermometer in the introduction to his compendium (Fearnley, 1865). Also as late as August 1887 the Pistor thermometer was situated at the northern wall according to recently discovered metadata (cf. Section 5). It is therefore possible that the thermometer at the eastern wall came out of use in 1894. The influence on the measurements, if the eastern thermometer was abandoned, might be as follows:

The sun was under the horizon for several hours for most of the year, before it came to the 2100 UTC observation. Even in midsummer the sunlight was shadowed by the local hills already before this last observation time of the day. But in summer some heat stored from earlier in the day must have been left in the concrete building's northern and western walls. Thus, at the 2100 UTC observation the temperature in sunny days could be higher at the western and northern thermometers than at the eastern one. Observations even at the northern wall might not have looked alarming for the observers as they also on that wall could observe temperature from thermometer under full shadow. However, the most preferable thermometer for observation was at the eastern wall, which had been in shadow since midday (Table 3). So, if the eastern thermometer came more or less out of use in 1895, it could have imposed an inhomogeneity on the series.

Table 3. The Oslo II series consists of observations from three windows of the Astronomical Observatory building, on the eastern wall (E), the western (W), and the northern (N). For each observation time there is 'preferable walls' that have been in shadow for several hours. For 'the possible walls' the thermometers are in shadow, but the walls may have been heated shortly before the observation time by direct sunlight.

Observation times (UTC)	06	08	13	15	21
Dec. preferable	W, N, E	W, N	N, E	N, E	E, N, W
Dec. possible	W, N, E	W, N, E	N, E	N, E	E, N, W
June, preferable	W	W	N	N	E
June, possible	W, N	W, N	N, E	N, E	E, N, W

Hypothesis II: CET was introduced in Norway from 1 January 1895. But this did not imply changes for all Norwegian meteorological stations as most of them continued to follow local time (Mohn, 1910, p. VII). It was not before 1920 that all stations in the network of MET Norway followed CET (Norw. Met. Inst., 1920, pp. VI & XIV). What the astronomers did is not known to us. In their observational practice it is evident that they had a conservative attitude. They continued their series with window thermometers in the same way as they started in 1837 without specifying which time reference system was used, so it is not obvious that they changed to CET in 1895.

The question now is which of the hypothesis can be verified, if any, and which can be falsified, if any. The hypotheses do not exclude each other. Hypothesis I seems reliable in the way that it quantitatively explains the seasonal variation of the corrections. The warming of the western and northern walls must be largest in spring and summer, when the day is longest and solar radiation at its highest. The problem with Hypothesis I appears with testing against neighbouring stations. Adding the corrections from the internal testing makes the Oslo II series inhomogenous, without corrections the series is homogenous.

If the Observatory changed from Christiania time to CET in 1895, according to Hypothesis II, they would have observed 17 min earlier than they had done in the years 1837–1894. On average the 2100 UTC observation should be warmer after the change. The hourly dataset from Blindern (1992.12–2009.06) with linear interpolation between neighbouring hours were used for calculation of the temperatures 17 min earlier than CET. As expected the temperature difference for the 2100 UTC (Table 4, line 21^{h}T) was largest in MJJ, about 0.3 °C, whereas the difference at the darkest months was zero. The value for MJJ was, however, lower than detected by internal testing (see Section 4.3), but with lower morning temperatures due to earlier observations also the reference for the testing (mT = mean temperature of observations at 0600, 0800, 1300, 1500) was changed. Taking the difference $\Delta T = 21^{h}\mathrm{T} - \mathrm{mT}$ gives

© 2014 Royal Meteorological Society

10970088, 2015, 12, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4223 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 4. The temperature difference (ΔT, °C) between the observation at 21 hour (21^{h}T) and the mean (mT) of the other observations (0600, 0800, 1300, 1500 UTC) caused by an overall change of observations times by −17 min. The Oslo−Blindern hourly dataset (1992.12−2006.06) is used.

	January	February	March	April	May	June	July	August	September	October	November	December
mT	0.00	−0.04	−0.10	−0.15	−0.19	−0.17	−0.17	−0.17	−0.11	−0.04	0.00	0.01
21^{h}T	0.02	0.06	0.12	0.18	0.27	0.30	0.29	0.22	0.12	0.07	0.04	0.02
ΔT	0.02	0.09	0.22	0.34	0.47	0.48	0.46	0.39	0.23	0.11	0.03	0.00

values close to those from the internal testing (Table 4). By use of Equation (A6a) in Appendix A1, it was revealed that the warmer 2100 UTC observation was compensated for by colder 600, 800 and 1300 observations, whereas 1500 UTC was not much changed (not shown). However, an overall changed observation time of 17 min had no effect on the calculated mean temperature. This fully agrees with the testing results against the neighbouring stations. Thus, Hypothesis II was accepted as it is in accordance with the fingerprints of this change, that is, strong seasonality and no effect on mean temperature. This finding also requires further interest as it shows that internal testing can be a powerful tool for detecting changed observation times even with changes as small as 17 min.

At this stage, the only correction added to the 18651 Oslo II series is for the period 1837−1920 for mercury droplets in the glass tubes, whereas the inhomogeneity of 1894 was not adjusted for as it did not affect in the monthly mean values, which is the subject for this article. It remains, however, to decide which stations should be included in the composite Oslo series: 18651 Oslo II or 18650 Oslo I.

4.5. Comparison of 18651 Oslo II with 18650 Oslo I

These series overlap through a period of 57 years, from 1877 to 1933. They were both located at the same place, at the Astronomical Observatory, but their radiation screens were fundamentally different. Whereas 18651 Oslo II used the Observatory house as radiation screen, 18650 Oslo I used a freestanding Wild screen (cf. Station history). Also the observation hours were different except that of 1300 UTC. In order to test the quality of the radiation screening, the daily values of the two series for 1300 UTC were compared by grouping into four seasons and five classes of cloud cover (Figure 6).

It is evident that there were large, systematic differences between the classes. For cloudy weather there were no differences in any season, while differences systematically became larger with decreasing cloud cover in all seasons. The effect was smallest in winter (about 0.5 °C for zero clouds, whereas in summer there were differences as large as 2.5 °C, showing that the series Oslo I was much warmer than Oslo II. Consequently the Wild screen must have been a poorer radiation screen than the observatory building, so the thermometer in the Wild screen must have had higher temperature than the ambient air outside the screen. We may say that the thermometer in the Wild screen was overheated. Such 'overheating problems' were recognized and debated already in the late 19th century (Middleton, 1966,

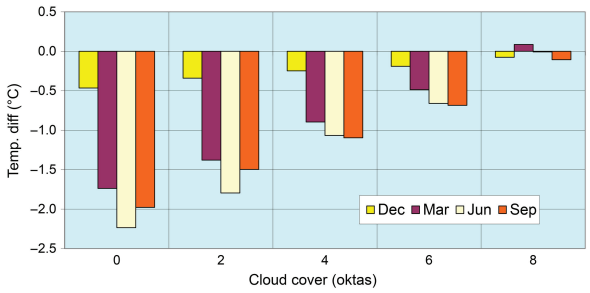


Figure 6. Difference between the observations of 18651 Oslo II and 18650 Oslo I for the observation hour 1300 UTC, grouped by season and cloud cover class for station 18651 Oslo II.

pp. 227−233) and they still are (Nordli *et al.*, 1997). The fact that the Observatory thermometers were not sheltered from precipitation may have cooled them by evaporation if wet by rain showers, so they might have been biased too low in those situations. But this cannot explain the much higher temperatures in the Wild screen than at the windows of the Observatory, as the largest differences occur under clear sky (Figure 6). The 18650 Oslo I series was also found to be inhomogenous by *HomeR*, see supporting material, Appendix S1.

The overheating and inhomogeneity problems of the 18650 Oslo I series are good reasons for not using it in the composite Oslo series and relay on 18651 Oslo II instead. Moreover the Oslo II series is much longer than Oslo I.

4.6. Homogeneity testing of 18700 Oslo−Blindern 1937−2012 using HomeR software

The series Oslo−Blindern is the only long-term series in Oslo that is still running (Figure 4). The Blindern area is near downtown Oslo within the domain of the Oslo fjord, which creates a local atmospheric circulation cell during winter as most of the fjord has open water. In midwinter during clear skies the temperatures at Blindern are higher than at many potential reference stations inland. Stations in the area of the fjord were thus preferred in the network of series for testing, but also inland stations were alternatively used. An additional difficulty, however, was that two of the suitable reference stations near the fjord, Rygge and Fornebu, are or have been on airports, where many changes have taken place, so it was not surprisingly that many inhomogeneities were detected by *HomeR* for those series. The network that was finally used for the correction of Oslo−Blindern is shown in the supporting material (Table S1), and the corrections themselves are shown in Table 5, columns marked Blindern.

© 2014 Royal Meteorological Society

10970088, 2015, 12, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4223 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 5. Adjustments for the Oslo composite series added to the raw data of the original series: 18650 Oslo I, 18651 Oslo II, 18700 Oslo–Blindern. In the period 1837–1920 an instrument correction of 0.3 °C is added in all months.

Period	1837–1920	1921–1933	1934–1937.02	1937.03–1980	1981–1995	1995–2012
Raw data	Oslo II	Oslo II	Oslo I	Blindern	Blindern	Blindern
January	0.21	−0.09	−0.13	0.15	0.16	0.00
February	0.30	−0.00	−0.40	0.17	0.13	0.00
March	0.16	−0.14	−0.60	0.05	0.14	0.00
April	0.08	−0.22	−0.60	0.03	0.16	0.00
May	0.00	−0.30	−0.70	0.02	0.22	0.00
June	−0.02	−0.32	−0.47	0.12	0.29	0.00
July	0.04	−0.26	−0.47	0.18	0.37	0.00
August	0.06	−0.24	−0.20	0.17	0.31	0.00
September	0.06	−0.24	−0.27	0.05	0.19	0.00
October	−0.05	−0.35	−0.57	0.00	0.10	0.00
November	−0.07	−0.37	−0.67	0.01	0.12	0.00
December	0.08	−0.22	−0.80	0.14	0.17	0.00
Year	**0.08**	**−0.22**	**−0.49**	**0.10**	**0.19**	**0.00**

All of the corrections of Oslo–Blindern were positive, but rather small at least for the annual values. The oldest data from 1937 to 1980 were corrected by only +0.1 °C. One possible cause might be the local urbanization that took place at Blindern in the 1970s by the expansion of the University, leading to higher temperatures in the Blindern area. The corrections in the more recent period 1981–1995 had a marked seasonal variation. Also for these corrections we had no direct metadata support, but deciduous trees and hedges and a few conifers in the vicinity (cf. Figure 5) have grown and been cut through the history of measurements, in particular to the west of the observational site. If trees were cut in the southern and western sectors around 1995, making the temperature screen more exposed to sunshine, it explains why the largest corrections were seen during summer.

4.7. Joining the series 18651 Oslo II and 18700 Oslo–Blindern using HomeR software

A homogenized series existed at both sides of the large relocation in 1937 after applying the corrections to Oslo II (Sections 4.2 and 4.4) and to Oslo–Blindern (Section 4.6). However, coupling them together may still lead to an inhomogenous composite series. The composite was tested by the *HomeR* software within the time interval 1890–1980, that is, about 44 years on each side of the relocation as the months 1934.01–1937.02 are missing. The expected inhomogeneity caused by the relocation was detected and corrected for (the corrections are shown in Table 5, whereas the network used is provided by the supporting material, Appendix S1). The *HomeR* software also has a feature for interpolation (Mestre *et al.*, 2013, p. 55), which could have been used directly for filling the missing months, 1934.01–1937.02, between the two long series. However, these interpolations were based on series outside Oslo, whereas other observations also did exist in Oslo (the Oslo I station). The challenge was to fill in the gaps by using data from Oslo I at the same time as the homogeneity of the composite series was kept. How this was done is described in Appendix A5.

The relocation in 1937 not only involved a 3 km horizontal distance, but also increased the altitude of the measurements from 25 to 94 m a.s.l. As expected, the corrections were negative for all months due to increased altitude, but the values differed in magnitude throughout the year (Table 5 column 3). The absolute values were less during late winter (even zero in February) when inversions may occur rather frequently. In spring and early summer, when the fjord was still cold compared to the land areas and inversions not so pronounced as in winter due to stronger short wave radiation, the temperatures were lower. During late autumn (October, November) clouds were more frequent over Oslo and consequently inversions were less frequent, thus the corrections were large. The resulting annual corrections of −0.22 °C implied an average temperature gradient of only -0.32 °C per 100 m increase of altitude, which shows that inversions plaid an important role in the Oslo climate.

In summary, the Oslo composite series 1837.04–2012.12 consists of only two principal series, 18651 Oslo II (1837.04–1933) and 18700 Oslo–Blindern (1937.03–2012.12), though the months 1934.01–1937.02 were filled in by adjusted values from the series 18650 Oslo I. The entire homogenized, composite temperature series is for brevity named *The Oslo series*, which will be subject to the following temperature analysis.

4.8. The long-term temperature evolution

The Oslo temperature series starts in a period when anthropogenic influence on climate was very low. In Oslo, where a limited industrialization took place along the river Akerselva, far from the Observatory site (ca. 3 km), the main source of emissions was the use of coal, gas and firewood for food preparation, heating and light in homes, offices and streets. The temperature in Oslo increased during the last 2/3 of the 19th century, although insignificantly according to linear regression analysis (Table 6). It continued to increase through the 20th century. The largest increase was in the latest 50 years of the series, in particular for winter: 0.55 °C per

© 2014 Royal Meteorological Society

10970088, 2015, 12, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4223 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 6. Linear trends (°C) of the Oslo series (1837.04–2012.12) grouped by season, given as temperature increase (decrease) in specified periods (after the tilde mark corresponding trends per decade).

Seasons	The 19th century: start–1900	The 20th century 1901–2000	Last 50 years 1963–2012	Obs. period start–2012
Annual	0.6 ~ 0.09	0.6 ~ 0.06*	1.7 ~ 0.34*	1.5 ~ 0.09*
Winter	1.6 ~ 0.25	0.2 ~ 0.02	2.7 ~ 0.55*	1.8 ~ 0.11*
Spring	0.5 ~ 0.07	1.0 ~ 0.10*	2.1 ~ 0.43*	2.0 ~ 0.12*
Summer	0.3 ~ 0.04	0.5 ~ 0.05	0.8 ~ 0.16	0.8 ~ 0.05*
Autumn	−0.1 ~ −0.00	0.9 ~ 0.09*	1.3 ~ 0.27*	1.4 ~ 0.08*

*Significant values, $p \leq 0.05$, calculated by SPSS statistical package No. 21: http://pic.dhe.ibm.com/infocenter/spssstat/v22r0m0/index.jsp, see also (Wilks, 1995, pp. 168–169).

Table 7. Highest and lowest seasonal and annual mean values (°C) for the Oslo series 1837.04–2012

	Winter	Spring	Summer	Autumn	Annual
Highest value	1.0 ~ 1989	8.0 ~ 1990	18.9 ~ 1947	9.2 ~ 2006	7.8 ~ 1990
Lowest value	−8.3 ~ 1871	1.3 ~ 1867	13.6 ~ 1851	3.7 ~ 1915	3.6 ~ 1838

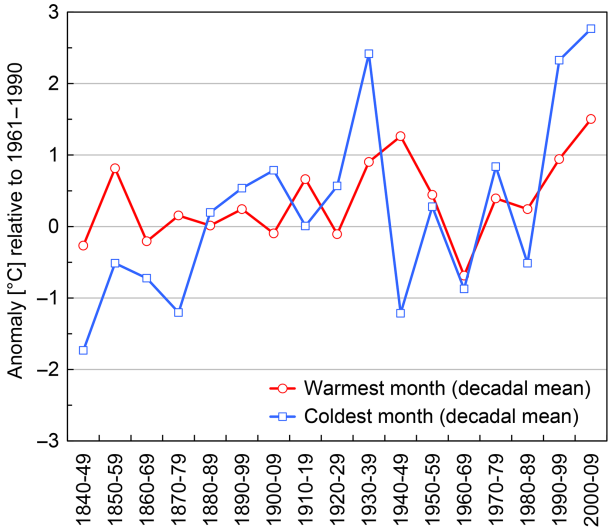


Figure 7. Time evolution of the decadal mean temperature anomalies of the warmest (circle) and coldest (square) months of each year. Anomalies are relative to the standard normal period 1961–1990.

decade. The temperature increased significantly over all seasons, but with very different amounts in different seasons. The temperature increase in summer for the whole observational period 1837–2012, was less than a half of that in winter and spring, which were the seasons with largest increase. The annually mean temperature increased by 1.5 °C during the 175 years of the period of observation.

Long-term warming trends in the series are also reflected in the timing of the coldest and warmest years and seasons in the series (Table 7). The highest mean seasonal temperatures are all from the period after 1946, whereas the lowest temperatures occurred before 1916. For annual mean temperature, the warmest year was 1990, and the coldest year 1838, the very first complete year in the series.

It was also revealed that the decadal mean temperatures of the coldest months in each year has increased more than the warmest one (Figure 7) in accordance with the stronger warming in winter than in summer. The increases for the coldest and warmest months were found to be 0.13 °C and 0.06 °C per decade respectively using linear regression. Decades with particularly cold 'coldest months' were found in the 1840s, 1870s and 1940s, whereas the 1960s had the lowest mean of the warmest months. The highest values were found for the last decade for both series.

Temperature variations were also studied by wavelet analysis, which not surprisingly revealed that the most power lied in the annual variations (not shown). With those variations removed the most interesting findings were significant variations on timescales of 10–20 years (Figure 8), but not for the whole series. The pronounced 10–20 timescales were only found after The Early 20th Century Warming (E20thCW) (e.g. Johannessen *et al.*, 2004), which was a rapid warming at high latitude (at least north of 60°N in the Atlantic sector). In south-eastern Norway, it culminated in the 1930s (Figure 9(a)). In the 19th century, most power was found on time scales of about five years, see Section 5.

The Oslo series was also smoothed with a Gaussian filter visualizing non-linear long-term variations in the series (Figure 9). The E20thCW event is pronounced in all seasons (except spring) as temperature increased rather abruptly from the 1920s up to a local maximum in the 1930s. The annual temperature remained high also during the two following decades. Thereafter temperature decreased and reached a local minimum in the 1960s (except autumn), before a strong increase, although with the expected decadal to bi-decadal variations seen in the wavelet analysis, set in during the latest 50 years. Both the annual and the seasonal series show high temperatures at the end of the series, in particular the latest 15 years, although in winter two rather cold years occurred (2010 and 2011).

Best-fit linear trends of the Oslo series were compared with corresponding trends from some long-term series in south-eastern Norway for the period 1874–2011 (Table 8). Only small differences can be seen in the table; for Oslo the trend for annual values amounts to about 0.9 °C per 100 years. Three series Oslo, Færder (Andresen, 2011) and Ås (Gjelten, 2011) are stations so close to each other that the networks used for homogenization to a certain extent consisted of the same stations. For Dombås the situation was different as the station is located far inland, two degrees latitude further north than Oslo, but still the annual trend did not differ from those near Oslo.

Compared with some homogenized long-term series from abroad, the trends of the Oslo and Stockholm series in the period 1838–2010 were almost equal when extra corrections were applied to the Stockholm series for spring and summer in the period 1838–1858, as taken from the

© 2014 Royal Meteorological Society

10970088, 2015, 12, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4223 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

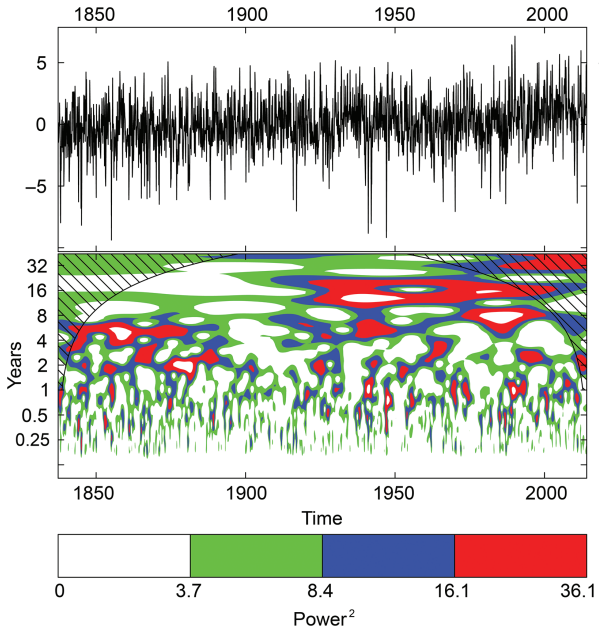


Figure 8. The upper panel shows the anomalies (°C), the time series from which the annual cycle has been subtracted. The lower panel shows the results of the wavelet analysis, with different colouring indicating different power levels; the time runs along the *x*-axis and the *y*-axis represents different time scales (periodicities in units of years). The green, blue and red shading indicate high power, and can be found at the levels associated with ∼5 year time scales in the early part, although there is also some indication of a brief presence of shorter time scales too (∼2–3 year time scales at around 1870). The dominant time scales towards the end of the period are found to be associated with 10–20 years, but there is also an ephemeral period with more high-frequent variability during the 1990s. The hatched regions show the cone of uncertainty near the end points of the time series where the wavelet analysis returns less reliable results to various end-effects.

Bolin Centre Database (http://bolin.su.se/data/stockholm/; last accessed 31 October 2014), Stockholm University (cf. Table 8, 'Stockholm2'). For the same period the trend for Central England series was less, 0.6 °C (http://www.metoffice.gov.uk/hadobs/hadcet/data/download.html; last accessed 31 October 2014), but for St Petersburg higher, 1.4 °C (http://aisori.meteo.ru/ClimateE; last accessed 31 October 2014). The extra corrections for the 'Stockholm2' series were the same as used in Moberg *et al.* (2005) in agreement with findings in Moberg *et al.* (2003), see Section 5.

5. Discussion

The findings of the present study have led to changes in our opinions of which data sources should be included in the long-term homogenized Oslo series. Historically all the old Oslo observations conducted by MET Norway have been considered as one series (Harbitz, 1963, p. 108). This means that the stations 18652 Oslo–Kirkegata, 18653 Oslo–Nordahl Bruunsgate and 18650 Oslo I have all been stored in MET Norway's data base (http://www.eklima.no; last accessed 31 October 2014) under the same identifier, 18650, and the same name, Oslo I. Due to the present analysis, this convention is changed, so the identifiers and names are in accordance with Table 1. The homogenized series is now stored under 'Homogenized monthly values' with the same identifier, 18700, as the last station in the series, Blindern. This signals that the homogenized Oslo series is homogenized so that new monthly values from Blindern can be added to the series without corrections. Our 'philosophy' has been to join the longest and most reliable series disregarding institutional borders. In consequence, MET Norway's own observations in the period 1866–1933 are now excluded, while observations conducted by the Astronomical Observatory included. It is common that homogenized series are changed when new homogenizations are performed, but rather uncommon that much of the raw data sources in a homogenized series are changed.

The formula for calculating the mean temperature for Oslo II was revisited based on a new hourly dataset from Blindern as early calculations had been made with very limited datasets. The result was very satisfying as the RMSE was less than 0.1 °C and no biases were detected.

A major inhomogeneity in 18651 the Oslo II series is of great concern. Homogeneity testing pointed out 1920 as break year, which is also in accordance with metadata. Under recalibration of the two Greiner thermometers in February 1921, it was detected that droplets had been released from the mercury columns, leading to an instrument correction of +0.3 °C for both of them. The observations continued with the same thermometers so either the droplets must have been successfully joined to the mercury strings, or alternatively, the instrument correction had been added to the observations directly by the observers or later in the written protocols. We solved the 1920 inhomogeneity problem by adding the instrument correction to the Oslo series for the period 1837–1920. However, the problem with this is twofold. We do not know for sure that the droplets were in the thermometers already from the beginning in 1837, and we know that the Greiner thermometers were only two of the three thermometers that have been used from the beginning. Birkeland (1926, p. 18) discussed this potential source for inhomogeneity in-depth, stating that the droplets were present in the tube at temperatures 45–50 °C. He also mentioned an account from 23 August 1887 telling about droplets in the tube, and further stated for sure that the droplets had not been released by strokes. He further hypothesized that the droplets had been released in connection with a boiling point test, when the thermometers have been left in horizontal position for cooling. There is support for this in Hansteen's 'correction book', which shows that the final test was just a boiling point test.

However, under the final phase of this study important accounts on the thermometers were discovered (cf. supporting material, Appendix S2) in the handwritten protocol for 1887 (Norw. Met. Inst., 1887) from which digitalization was performed. The accounts tell that Greiner I, Greiner II and Pistor are placed at the western, eastern and northern window respectively. According to (Birkeland, 1926, p. 18) Greiner I and Greiner II were in 1923 at the western and northern windows. A third thermometer

© 2014 Royal Meteorological Society

10970088, 2015, 12, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4223 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

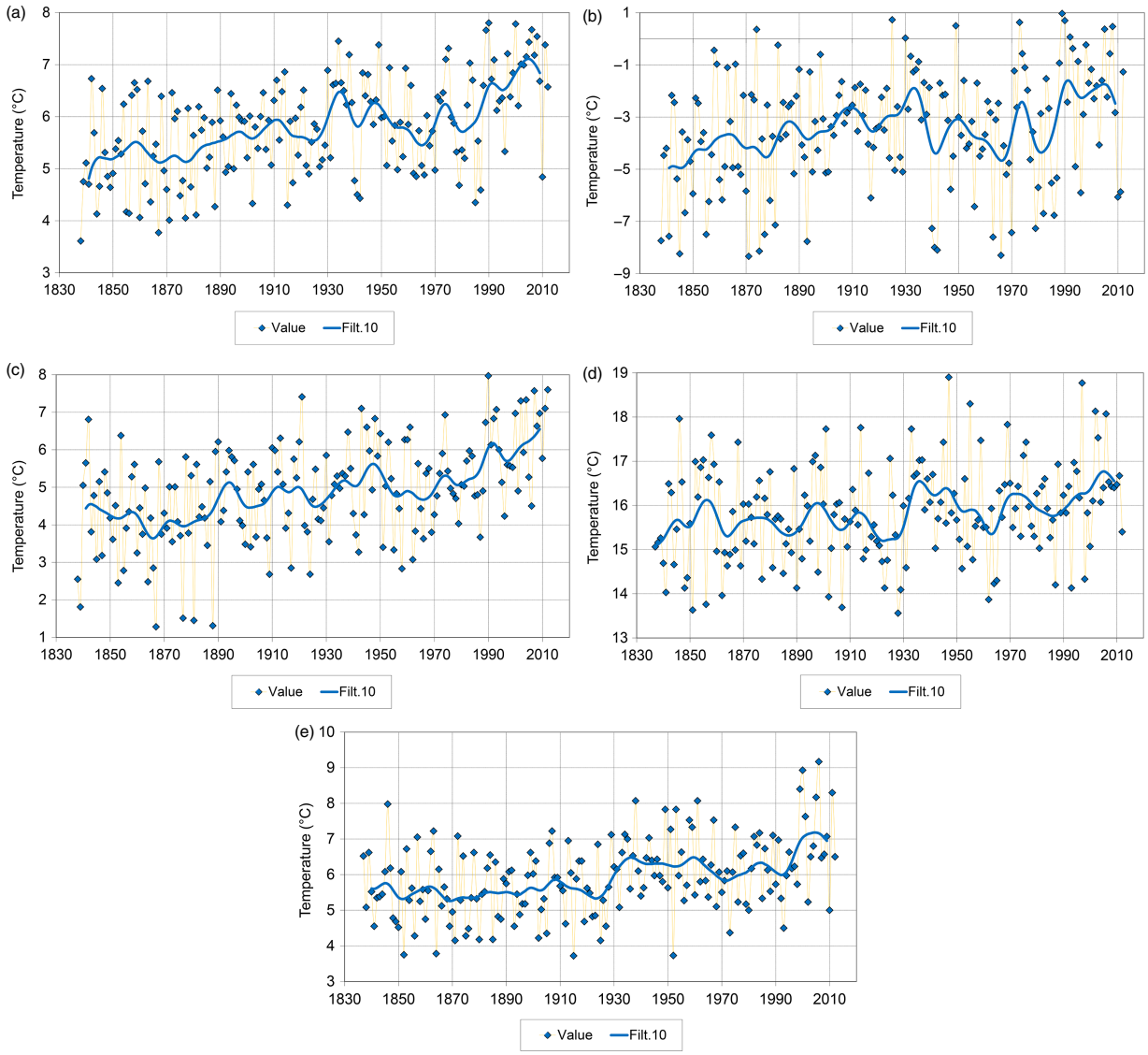


Figure 9. Annual and seasonal means of the Oslo series (symbols), and Gaussian filter with standard deviation 3 years in the Gaussian distribution, corresponding to about 10 year rectangular filter for (a) Annual, (b) Winter, (c) Spring, (d) Summer and (e) Autumn values.

(not specified) was at the eastern window, but was seldom in use. In agreement with Birkeland the accounts from 1887 tell that the Greiner I and Greiner II had small mercury droplets in their tubes. The new information is that the droplets were joined to the mercury columns. Comparison with the Pistor thermometer before and after the joining showed that the size of the droplets amounted to +0.26 °C for Greiner I and +0.49 °C for Greiner II. For Greiner I the droplet was approximately of the same size as seen in 1921, whereas the droplet in Greiner II was larger in 1887 than in 1921. An explanation for the forming of the droplets might be that mercury has slowly evaporated from the columns when heated by direct sunshine, and that condensation had occurred at the end of the tubes. It is revealed that the instrument corrections of the Greiner thermometers have not been as stable through the station history as suggested by Birkeland (1926, p. 18). The data from before 1921 need positive corrections but with our limited knowledge of the thermometers and their use, it is impossible to do this in detail. In agreement with the homogeneity testing and for simplicity we have decided to apply a correction of 0.3 °C back to the start of the series in 1837.

A network of stations established in the 1870s made it feasible to use HomeR for homogeneity testing, see Methods. The terminology of *HomeR* was adopted so with networks of stations we mean different selections of neighbouring stations. If one or more neighbouring stations are replaced by other stations we say that the network is changed. The reliability and robustness of the results were tested by changing networks used in the program, and also if appropriate, with different decisions taken interactively when running the program. A final evaluation was undertaken based on how well the potential shifts were able to match metadata (http://www.eklima.no; last accessed 31 October 2014), not only for the Oslo series, but for all series in the networks. The networks that seem to give the most reliable results are listed in the supporting material (Table S1), and from those program runs the final

© 2014 Royal Meteorological Society

10970088, 2015, 12, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4223 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 8. Centennial (per 100 years) linear temperature increase with standard deviation for the Oslo series compared with a few other homogenized long-term series. Two versions of the Stockholm series were downloaded from Stockholm University, Bolin Centre Database. For 'Stockholm 2' extra corrections were applied compared to 'Stockholm 1' for the period 1756–1858 for the months May to August, −0.3 °C for May and August, and −0.7 °C for June and July. All trends were found to be significantly ($p \leq 0.05$) different from zero except 'Stockholm 1' for summer.

Period	Series	Annual	Winter	Spring	Summer	Autumn
1838–2010	Central England	0.6 ± 0.1	0.5 ± 0.2	0.6 ± 0.1	0.5 ± 0.1	0.9 ± 0.1
	Oslo	0.9 ± 0.1	1.1 ± 0.3	1.1 ± 0.2	0.5 ± 0.2	0.8 ± 0.1
	Stockholm 1	0.7 ± 0.1	0.8 ± 0.3	1.1 ± 0.2	0.0 ± 0.2	0.8 ± 0.2
	Stockholm 2	0.8 ± 0.1	0.8 ± 0.3	1.3 ± 0.2	0.5 ± 0.2	0.8 ± 0.2
	St. Petersburg	1.4 ± 0.2	1.7 ± 0.4	2.1 ± 0.2	0.9 ± 0.2	1.0 ± 0.2
1874–2011	Oslo	0.9 ± 0.2	0.7 ± 0.5	1.3 ± 0.2	0.6 ± 0.2	1.0 ± 0.2
	Ås	1.0 ± 0.2	1.0 ± 0.5	1.4 ± 0.3	0.9 ± 0.2	0.9 ± 0.2
	Dombås	0.9 ± 0.2	0.8 ± 0.5	1.3 ± 0.3	0.5 ± 0.2	1.0 ± 0.3
	Færder Fyr	0.9 ± 0.2	0.8 ± 0.4	1.2 ± 0.3	0.6 ± 0.2	1.0 ± 0.2

corrections were also achieved. Examples of practical use of *HomeR* seem very limited, certainly due to its novelty. One example is from Mestre *et al.* (2013) dealing with a network of stations surrounding Vienna. Our focus has been only Oslo, so our choice of *HomeR* was thus taken entirely because its powerfulness, not because of an intention of homogenizing a network.

In 1837 the Observatory had a rural position 2 km from the centre of the town, but in the following years the town expanded, and after the close down of the Observatory, the expansion came to its neighbouring areas, the nearest house only about 40 m from the eastern wall of the observatory building. But as long as the meteorological observations were carried out (to the end of 1933), the Observatory had a free-lying position so that an urban heat island effect is not expected. Moreover, the homogeneity testing and corrections were performed relative to neighbouring stations giving no indication of urban influence. After the relocation in 1937, the instruments were moved to another free position that remained open for many years. At a distance of about 300 m the nearest buildings of the Oslo University came in the 1960s and 1970s. Again the series was tested against rural neighbouring stations that led to the detection of two small breaks in the series. Positive corrections in the first part of the series reduce the trend. Therefore, a small urban heat island effect may be part of the explanation, but the break is mainly in the summer months, so more likely it is due to changed vegetation to the south and west of the measuring point.

If an urban heat island effect still is present in the Oslo series after the homogenization, one should expect its trend to be higher than for the rural neighbouring series during the period of urban expansion. For the period 1874–2012 this is not the case (Table 6). However, in recent years, particularly in the period 2009–2013, new buildings came in the western sector of the station, the nearest one only 100 m SW of the site. It is possible, or even likely, that the Oslo series will get an urban heat island effect in the future.

Thermometers fastened to the northern wall of houses or outside windows were common in the 18th, 19th and even in the early 20th century. This could lead to a warm bias in the summer morning or evening observations if the thermometers were insufficient protected by sunshine, which is of importance even for global datasets (Hartmann *et al.*, 2013, p. 198). For the Stockholm series this problem has been carefully investigated by Moberg *et al.* (2003) by multiple regression analysis with temperature as predictand and several weather elements as predictors. For the JJA average, a model that included cloud cover, meridional wind speed and local air pressure, together explained 65% of the temperature variance. Of these elements cloud cover was found to be the far most important one. By using temperature estimated by this regression model as reference series in a homogeneity test, a warm bias of the observed summer temperatures was estimated to be 0.7–0.8 °C although not accurately determined. A more direct approach for solving the bias problem has been performed by Böhm *et al.* (2010), who used parallel measurements of historical and modern thermometer exposure at one station in Austria. Based on these measurements a correction model was deduced, which was applied on data from other stations. Warm biases were found to be largest during June, estimated to 0.21–0.93 °C for stations in an area around the Alps called the Greater Alp Region (GAR). The biases depended on the site of the instrument and formula used for calculation of monthly mean temperature. Moberg *et al.* (2003) recommend 'an assessment of how widespread the problem is with inefficient radiation screening in early temperature series', not only for a further 'assessment of Stockholm and Uppsala temperatures, but because this would be necessary for more reliable reconstruction of climate change in the whole of Europe back to the 18th century'. In this context the 18651 Oslo II is of particular interest, as three walls were in use for the observations, so that proper shadow of at least one thermometer was always available for the observers. Comparing the trends in the Oslo and Stockholm series is just another independent indication that a correction of the Stockholm series is needed. These series are comparable also because urban trends are removed from both, whereas the steeper trend in the St. Petersburg series may at least partly be urban. This is in agreement with the findings in Jones and Lister (2002), see Table 8.

© 2014 Royal Meteorological Society

10970088, 2015, 12, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4223 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

The time series presented in Figure 9 show that the coldest seasons and years were generally found in the early part of the record, as all seasons have experienced a warming trend. Some of the coldest winters, however, can be found in 1940–1970 when the average temperature was around 2° lower than the more recent period 1990–2010 and comparable to the 1850–1900. The autumn temperature had a negligible trend before 1925 and during 1935–1995 (Figure 9(e)) with one of the coldest autumn seasons in 1952. The different epochs with colder and warmer conditions reflect natural variations, and it is possible to deduce directly from the time series of annual values (Figure 9(a)) that there were stronger variations from one year to the next in the early part compared to the recent times where the temperature could persist with similar magnitude for several years. A more objective approach to study the persistence and the nature of natural variations, however, can be made through a wavelet analysis.

The wavelet analysis sheds some further light on the temperature record, which raise a new set of questions. One is whether the detected shifts in the spectral characteristics of the series were connected to the change of observational procedures and inputs in the long-term Oslo temperature series. The cessation of the ∼5 year time scale approximately coincided with the time when the Observatory changed the observation times from Christiania Time to Central European Time (1895), and the 10–20 time scale dominance started at around the time the measurements from the current location at Blindern started (1937). However, it is difficult to explain how such changes physically could affect these multi-annual time scales. Another question is whether this shift in frequency may be a real feature of the natural variability and suggests that the most dominant frequency has not been stable. One explanation may be that there exist different regimes in the weather that span over decades and that the abrupt shift in climate represented by the E20thCW is part of the story. We also performed wavelet analyses for the Central England, Stockholm and St. Petersburg series. For Stockholm and St. Petersburg we detected similar frequency shifts as in the Oslo series, suggesting that the shifts are real and present in a region stretching eastward from Oslo.

The Oslo series has been relocated only once, and at least until recently, its site had a semi-rural environment. The conditions for the Oslo II part are remarkable: the same location was used for 97 years, mainly the same thermometers for 85 years, and stable screening of the thermometers by three different walls, one of them always in shadow.

The composite record as well as the raw data series may be downloaded from the climate database of MET Norway: http://www.eklima.met.no; last accessed 31 October 2014.

6. Conclusions

A 175 years long homogenized record for Oslo of monthly mean temperatures was established mainly composed of two series. These are 18651 Oslo II and 18700 Oslo–Blindern. The 18651 Oslo II series, run by Astronomical Observatory (1837.04–1933.12), was found to be homogenous for mean temperature as far as testing was possible after correction for an instrumental bias. By use of internal testing it was revealed that the Observatory changed the observation times from Christiania Time to Central European Time in 1895, which resulted in an inhomogeneity during spring and summer for individual observation times, but this did not affect the monthly mean temperature.

The 18700 Oslo–Blindern series (1937 to present) at the premises of MET Norway was tested and it was found that two sets of monthly corrections were needed (1937–1980) and (1981–1995) in order to homogenize the record.

The two long-term series 18651 Oslo II and 18700 Oslo–Blindern were successfully coupled and corrected, so as to be included in a homogenized, composite Oslo series starting in 1837. To fill in a short gap, 1934.01–1937.02, data from 18650 Oslo I was corrected and included. A correction was needed partly because the Oslo I series with radiation protection of a Wild screen encountered serious overheating problems.

For the whole Oslo series of 175 years the linear temperature increase was estimated to be 1.5 °C, with the steepest trend in the latest 50 year of the series. The Early Twentieth Century Warming was prominent in the Oslo series. The warmest year was 1990 and the coldest one 1838, the very first full year in the series. The coldest months of each year had a significant positive trend (0.13 °C/decade), twice as high as the trend for the warmest months (0.06 °C/decade).

The Wavelet analysis of the Oslo series suggests that the most dominant frequency in the variations has not been stable; in the beginning of the record, the strongest variations were characterized by ∼5-year periodicity, whereas the most pronounced variations towards the end of the record were substantially slower (10–20 year periodicity). This change coincides with the Early Twentieth Century Warming.

Acknowledgements

The project HISTKLIM at MET Norway is acknowledged for having performed the digitalization and quality control of the historical observations from Oslo, which has made the present analysis possible. We want to express our gratitude to our two reviewers for helpful comments. The reviewer, Prof. Anders Moberg, is also very much acknowledged for his detailed suggestions that have significantly improved the article.

Appendix

Appendix A1. Details about estimating monthly mean temperature

Köppen's formula: Night-time observations were lacking for many stations in the traditional network of MET Norway, but this was compensated by taking minimum temperature into account when the monthly mean was calculated. For this purpose a method introduced by

© 2014 Royal Meteorological Society

10970088, 2015, 12, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4223 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Köppen (1888) was adopted by the institute (Birkeland, 1936; Nordli and Tveito, 2008). The method was based on the mean of three daily observations at fixed hours, which were equally weighted, and daily minimum temperature, which had to be weighted by a coefficient k, known as Köppen's constant. MET Norway is the only meteorological institute in the world that has adopted Köppen's formula (Köppen, 1888) for the regular calculation of monthly mean temperature when hourly observations are lacking. The reason for its limited use might be that the formula was vehemently attacked shortly after it was introduced (Leyst, 1892). Much later the formula was strongly defended by Birkeland (1936). These references above are all in German. To our knowledge, only two institute reports exist written in English where Köppen's formula are described (Høgåsen, 1993; Nordli and Tveito, 2008). By the present paper it appears in a peer reviewed English journal.

Monthly mean temperature, T, based on three daily observation times and daily minimum temperature may be written:

$$T = aT_{06} + aT_{12} + aT_{18} + kT_n \tag{A1}$$

where $T_{06}, T_{12}, T_{18,}$ are the monthly means of the morning, midday and evening observations respectively, and T_n is the monthly mean of the daily minimum temperature. The three observations at fixed hours with the two equal time intervals between them, were equally weighted, a, whereas k was the weight for minimum temperature. A constraint was introduced, that is, the sum of the weights should be 1,

$$3a + k = 1 \tag{A2}$$

For simplicity the monthly mean of the observations at fixed hours, T_f, is also introduced,

$$T_f = \frac{1}{3}\left(T_{06} + T_{12} + T_{18}\right) \tag{A3}$$

By use of formulas (A2) and (A3) in (A1) and rearranging, Köppen's formula in its most general form may be written

$$T = T_f - k\left(T_f - T_n\right) \tag{A4}$$

The coefficient, k, has to be calculated for each month using data from a time interval of some years when hourly observations are available. Then T, T_f and T_n are known so that k is the only unknown quantity in Equation (A4). In the Arctic during winter, when no daily temperature wave exist, $k = 0$, whereas in a continental climate during midsummer in south eastern Norway, k takes values between 0.20 and 0.25.

Føyn's formula: If daily minimum temperature is lacking a Köppen like formula attributed to Føyn (Nordli and Tveito, 2008) has been used at MET Norway for calculation of monthly means:

$$T = T_g + k_g\left(T_{12} - T_g\right) \tag{A5}$$

where T_g is the mean of the morning and evening observations and T_{12} is the midday observation. The coefficient k_g has to be calculated for each month by using an hourly dataset.

Hansteen's formula: For the station 18651 Oslo II, Hansteen (Hansteen and Fearnley, 1849; Fearnley, 1867; Birkeland, 1926) introduced his own formulas for monthly mean calculation, for five and four observations a day (A6a) and (A6b), respectively:

$$T = \frac{1}{48}\left(11T_{06} + 7T_{08} + 7T_{13} + 8T_{15} + 15T_{21}\right) + C_a \tag{A6a}$$

and

$$T = \frac{1}{48}\left(16T_{08} + 7T_{13} + 8T_{15} + 17T_{21}\right) + C_b \tag{A6b}$$

where T_i is the temperature at hour i and C_a and C_b are constants that have to be calculated for each month by using an hourly dataset. The weights in the formulas are the sums of the time intervals in hours from one observation to its two neighbours. For example in formula (A6a) the observation at 21 h is more heavily weighted than any of the other ones as the sum of the time interval to its neighbours is 6 h + 9 h = 15 h. By dividing by 48, it is ensured that the sum of the weights is equal to 1.

Appendix A2. Contribution to mean temperature of the 2100 UTC observation

A change in the temperature observed at 2100 UTC will lead to a change in the monthly mean value, so by Hansteen's formula:

$$\Delta_T = \frac{15}{48}\Delta_{21} \tag{A7}$$

where Δ_T is the change of the monthly mean temperature and Δ_{21} is the change of the 2100 UTC temperature. The assumption is made that there was no change of the average in the other observations.

Appendix A3. Details about homogeneity testing

For the homogeneity testing we have used different methods:

Multiple Linear Regression procedure (MLR) (Vincent, 1998; Ducré-Robitaille *et al.*, 2003) also called the Vincent test, uses successively four regression models. After the application of each model the residuals are analysed in order to examine its fit. A first model is used to determine whether or not the candidate series is homogeneous. If not, testing is continuing for the detection of a trend, for a step, and for a step with trends on each side of the step.

Standard Normal Homogeneity Test (SNHT) (Alexandersson, 1986; Alexandersson and Moberg, 1997; Ducré-Robitaille *et al.*, 2003) has been widely used for testing of both precipitation series and temperature series. The first version of the test (Alexandersson, 1986) had a one step change as the only possibility, whereas in the version of 1997 both double shifts and a trend were possible outcome of the test. At any year the significance of a potential break or trend segment is examined. If no significant break or trend occurs the series is considered homogeneous.

The *AnClim*, http://www.climahom.eu/software-solution/anclim, by Petr Štěpánek, Pavel Zahradníček and Petr

© 2014 Royal Meteorological Society

10970088, 2015, 12, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4223 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Skalák is a software package, where both *MLS* and *SNHT* are included. This software was used for internal testing, whereas another software package, *HomeR*, was used in the period when neighbouring stations around Oslo were available.

The *HomeR* software was used following sequence of recommendations by the authors of the program (Mestre and Anguilar, 2011):

1. Pairwise detection: Testing series against each other (Mestre *et al.*, 2013, pp. 49–50);
2. Joint detection: Joining the results of pairwise tetection (Mestre *et al.*, 2013, p. 50);
3. Correction for inhomogeneities based on (1) and (2). By running the program interactively metadata was taken into account (Mestre *et al.*, 2013, pp. 57–58);
4. ACMANT detection (Applied Caussinus Mestre Algorithm for homogenizing Networks of Temperature series) (Domonkos, 2011, 2013; Mestre *et al.*, 2013, pp. 56–57);
5. Further corrections for inhomogeneities;
6. Pairwise detection: As (1) but this time detection was performed on the corrected series;
7. Joint detection: As (2) but this time detection was performed on the corrected series;
8. Further corrections for inhomogeneities if necessary;
9. ACMANT detection based on input data;
10. Assessing the month of change;
11. Final corrections.

HomeR can be run interactively, which means that the user can change the decisions taken by the program while running, helping the program to take the right decisions, No. 3 above. If, for example, one change point (an inhomogeneity) occurs one or two years off the event that might have caused it according to metadata, the change point can be moved by the user to the correct year. In the next step the program will take the change into account. In *HomeR* features from other programs are also incorporated, that is, some innovations from ACMANT (Domonkos, 2011, 2013 and cleaning for outliers by the software CLIMATOL (Guijarro, 2013, p. 4). By the use of ACMANT, the month of the change was assessed (No. 10) and also the final monthly corrections (No. 11).

A4. Gaussian smoothing

Low passed filters may be used to 'smooth' the data series. One of these is a filter using the Gaussian distribution. By choosing a standard deviation, s, in the distribution, it is decided at what time scale the variations are smoothed. In the present article, we have chosen $s = 3$ in Equation (A8), which removes variations at smaller time scales than approximately 10 data points in the time series.

The filtered value at data point j is given by:

$$G_j = \frac{\sum_{i=1}^{n} w_{ij} x_i}{\sum_{i=1}^{n} w_{ij}}, \quad w_{ij} = \exp \frac{-1\,(i-j)^2}{2s^2} \qquad \text{(A8)}$$

s = standard deviation; w = the weight; n = No. of observations in the series; x = data point. A new series of filtered values is established by letting j run through all data points.

A5. Gap filling 1934.01–1937.02 for the composite Oslo series

The gap was filled using data from 18650 Oslo I, but adjusted to give the same long-term average as the corresponding values interpolated by *HomeR*. The modified values, z_{ij} for year i and month j, are given by Equation (A9):

$$z_{ij} = \left(\overline{y}_j - \overline{x}_j\right) + x_{ij} \qquad \text{(A9)}$$

$\overline{y}_j$ is the mean value interpolated by *HomeR* for the period 1934.01–1937.02 for month j. $\overline{x}_j$ is the mean value of 18650 Oslo I for the same period for month j. x_{ij} is the monthly mean of 18650 Oslo I for year i and month j.

Taking the mean values of Equation (A9) over the time interval 1934.01–1937.02 gives $\overline{z}_j = \overline{y}_j$, which means that the homogeneity of the composite series is not broken by the modifications.

Supporting Information

The following supporting information is available as part of the online article:

Appendix S1. The use of HomeR – documentation.

Appendix S2. Information on thermometers at 18651 Oslo II in the year 1887: Assessment of the size of the droplets in the tubes.

References

Aguilar E, Auer I, Brunet M, Peterson T, Wieringa J. 2003. Guidelines on climate metadata and homogenization (No. WCDMP-53 (WMO-TD No. 1186)). Paul Llansó, WMO: Geneva, Switzerland.

Alexandersson H. 1986. A homogeneity test applied to precipitation data. *J. Clim.* **6**: 661–675, doi: 10.1002/joc.3370060607.

Alexandersson H, Moberg A. 1997. Homogenization of Swedish temperature data. Part I: homogeneity test for linear trends. *Int. J. Clim.* **17**: 25–34, doi: 10.1002/(SICI)1097-0088(199701)17:1<25::AID-JOC103>3.0.CO;2-J.

Andresen L. 2011. *Homogenization of Monthly Long-Term Temperature Series of mainland Norway* (*met.no/Note No. 02/2011*). Norwegian Meteorological Institute: Oslo.

Auer I, Böhm R, Jurkovic A, Lipa W, Orlik A, Potzmann R, Schöner W, Ungersböck M, Matulla C, Briffa K, Jones P, Efthymiadis D, Brunetti M, Nanni T, Maugeri M, Mercalli L, Mestre O, Moisselin J-M, Begert M, Müller-Westermeier G, Kveton V, Bochnicek O, Stastny P, Lapin M, Szalai S, Szentimrey T, Cegnar T, Dolinar M, Gajic-Capka M, Zaninovic K, Majstorovic Z, Nieplova E. 2007. HISTALP – historical instrumental climatological surface time series of the Greater Alpine Region. *Int. J. Clim.* **27**: 17–46, doi: 10.1002/joc.1377.

Barlaup A. 1966. *Det norske meteorologiske institutt 1866-1966 (The Norwegian Meteorological Institute 1866-1966)*. Norwegian Meteorological Institute: Oslo.

Bergström H, Moberg A. 2002. Daily air temperature and pressure series for Uppsala (1722–1998). *Clim. Change* **53**: 213–252, doi: 10.1023/A:1014983229213.

Birkeland BJ. 1926. *Ältere Meteorologische Beobachtungen in Oslo (Kristiania). Luftdruck und Temperatur in 100 Jahren. (Old meteorological observations in Oslo. One hundred years of air pressure and temperature)*. Geofys. Publ. III, Geofys. Publ.: Oslo, 56 pp.

© 2014 Royal Meteorological Society

10970088, 2015, 12, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4223 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Birkeland BJ. 1930. *Ältere Meteorologische Beobachtungen in Bergen. Luftdruck und Temperatur seit 100 Jahren (Old meteorological Observations in Bergen. Air Pressure and temperature during 100 years)*. Geofys. Publ. V, Geofys. Publ.: Oslo, 56 pp.

Birkeland BJ. 1931. *Ältere Meteorologische Beobachtungen in Ullensvang. Luftdruck und Temperatur seit 100 Jahren (Old Meteorological Observations at Ullensvang during 100 years)*. Geofys. Publ. IX, Geofys. Publ.: Oslo, 40 pp.

Birkeland BJ. 1935. *Ältere Meteorologische Beobachtungen in Vardö. Luftdruck und Temperatur seit 100 Jahren (Old meteorological Observations in Vardø . Air Pressure and Temperature)*. Geofys. Publ. X, Geofys. Publ.: Oslo, 52 pp.

Birkeland BJ. 1936. *Mittel und Extreme der Lufttemperatur (Mean and Extreme of Air Temperature)*. Geofys. Publ. XIV, Geofys. Publ.: Oslo, 155 pp.

Birkeland BJ. 1949. *Old Meteorological Observations at Trondheim. Atmospheric Pressure and Temperature*. Geofys. Publ. XV, Geofys. Publ.: Oslo, 38 pp.

Böhm R, Auer I, Brunetti M, Maugeri M, Nanni T, Schöner W. 2001. Regional temperature variability in the European Alps: 1760–1998 from homogenized instrumental time series. *Int. J. Clim.* **21**: 1779–1801, doi: 10.1002/joc.689.

Böhm R, Jones PD, Hiebl J, Frank D, Brunetti M, Maugeri M. 2010. The early instrumental warm-bias: a solution for long central European temperature series 1760–2007. *Clim. Change* **101**: 41–67, doi: 10.1007/s10584-009-9649-4.

Camuffo D. 2002a. Calibration and instrumental errors in early measurements of air temperature. *Clim. Change* **53**: 297–329, doi: 10.1023/A:1014914707832.

Camuffo D. 2002b. Errors in early temperature series arising from changes in style of measuring time, sampling schedule and number of observations. *Clim. Change* **53**: 331–352, doi: 10.1023/A:1014962623762.

Caussinus H, Mestre O. 2004. Detection and correction of artificial shifts in climate series. *Appl. Stat.* **53**: 405–425.

Demarée GR, Lachaert P-J, Verhoeve T, Thoen E. 2002. The long-term daily central Belgium temperature (CBT) series (1767–1998) and early instrumental meteorological observations in Belgium. *Clim. Change* **53**: 269–293, doi: 10.1023/A:1014931211466.

Domonkos P. 2011. Adapted Caussinus-Mestre algorithm for networks of temperature series (ACMANT). *Int. J. Geosci.* **II**: 293–309.

Domonkos P. 2013. Measuring performances of homogenization methods. *IDŐJÁRÁS Q. J. Hung. Meteorol. Serv.* **117**, 91–112.

Ducré-Robitaille J-F, Vincent LA, Boulet G. 2003. Comparison of techniques for detection of discontinuities in temperature series. *Int. J. Clim.* **23**: 1087–1101, doi: 10.1002/joc.924.

Fearnley C. 1865. *Meteorologische Beobachtungen an der Koniglichen Universitäts-Sternwarte zu Christiania 1837–1863. (Meteorological Observations at the Royal University's Observatory in Christiania 1837–1863)*. H.J. Jensen: Oslo.

Fearnley C. 1867. *Meteorologische Beobachtungen aufgezeichnet auf Christiania's Observatorium, Complet 1837–1867 (Meteorological Observations at Christiania's Observatory, complete 1837–1867)*. H.J. Jensen: Oslo.

Gjelten HM. 2011. *Homogenitetstesting og analyse av temperaturserien på Ås (Homogeneity Testing and Analysis of the Temperature Series from Ås)*, Master thesis, No. 30 stp., 2011.

Guijarro JA. 2013. Climatological series shift test comparison on running windows. *IDOJARAS Q. J. Hung. Meteorol. Serv.* **117**: 35–112.

Hansteen C, Fearnley C. 1849. *Beschreibung und Lage der Universitäts-Sternwarte in Christiania. Auf Veranstaltung des Academischen Collegiums. (Description of the Astronomical Observatory in Christiania)*. P.T. Malling: Oslo.

Harbitz H. 1963. Oversikt over de offisielle meteorologiske stasjoner og observasjoner i Norge samt over rutine-bearbeidelsen av dem i årene 1866–1956. (The Official Meteorological Stations and Observations in Norway and routine publications in the years 1866–1956). Technical Report No. 6, Norwegian Meteorological Institute, Oslo.

Hartmann D, Klein Tank AMG, Rusticussi M. 2013. Observations: atmosphere and surface. In *Climate Change 2013*: *The Physical Science Basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change (IPCC), AR5*. IPCC: New York, NY.

Hesselberg T. 1917. *Meteorologien i Norge 50 aar. Festskrift (Meteorology in Norway 50 years)*. Oslo: Norwegian Meteorological Institute.

Høgåsen S. 1993. *Mean Temperature in Meteorology. (No. 40/93)*. DNMI-klima: Oslo.

Johannessen OM, Bengtsson L, Miles MW, Kuzmina SI, Semenov VA, Alekseev GV, Nagurnyi AP, Zakharov VF, Bobylev LP, Pettersson LH, Hasselmann K, Cattle HP. 2004. Arctic climate change: observed and modelled temperature and sea-ice variability. *Tellus* **56**: 328–341, doi: 10.1111/j.1600-0870.2004.00060.x.

Jones PD, Lister DH. 2002. The daily temperature record for St. Petersburg (1743–1996). *Clim. Change* **53**: 253–267, doi: 10.1023/A:1014918808741.

Köppen W. 1888. Über die ableitung wahrer Tagesmittel aus den Beobachtungen um 8 h a.m., 2 h p.m. und 8 h p.m. (On the calculation of true monthly means from the observations at 8 h a.m., 2 h p.m. and 8 h p.m.). *Ann. Hydrogr. Marit. Meteorol.* **16**: 341–354.

Leyst E. 1892. Die berechnung von Temperatur-mitteln aus Beobactungen zu den Terminen 8 Uhr VM., 2 Uhr und 8 Uhr MN (in German). (On the calculation of mean temperature from observations at the times 8 a.m., 2 p.m. and 8 p.m. CET). *Repert Für Meteorol* **15**: 1–35.

Manley G. 1953. The mean temperature of Central England, 1698 to 1952. *Q. J. R. Meteorol. Soc.* **79**: 242–261.

Manley G. 1974. Central England temperatures: monthly means 1659 to 1973. *Q. J. R. Meteorol. Soc.* **100**: 389–405.

Mestre O, Anguilar E. 2011. HOME_R Fast documentation Training School, 5-7 October 2011. University Rovira i Virgili, Tarragona, Spain.

Mestre O, Domonkos P, Pichard F, Auer I, Robin S, Lebarbier E, Böhm R, Anguilar E, Guijarro J, Vertachnik G, Klancar M, Dubuisson B, Stepanek P. 2013. HOMER : a homogenization software – methods and applications. *IDŐJÁRÁS Q. J. Hung. Meteorol. Serv.* **117**: 47–67.

Middleton WEK. 1966. *A History of the Thermometer and its Use in Meteorology*. The John Hopkin Press: Baltimore, MA.

Moberg A, Alexandersson H. 1997. Homogenization of Swedish temperature data. Part II: homogenized gridded air temperature compared with a subset of global gridded air temperature Since 1861. *Int. J. Clim.* **17**: 35–54, doi: 10.1002/(SICI)1097-0088(199701)17:1<35::AID-JOC104>3.0.CO;2-F.

Moberg A, Bergström H. 1997. Homogenization of Swedish temperature data. Part III: the long temperature records from Uppsala and Stockholm. *Int. J. Clim.* **17**: 667–699, doi: 10.1002/(SICI)1097-0088(19970615)17:7<667::AID-JOC115>3.0.CO;2-J.

Moberg A, Bergström H, Krigsman JR, Svanered O. 2002. Daily air temperature and pressure series for Stockholm (1756–1998). *Clim. Change* **53**: 171–212, doi: 10.1023/A:1014966724670.

Moberg A, Alexandersson H, Bergström H, Jones PD. 2003. Were southern Swedish summer temperatures before 1860 as warm as measured? *Int. J. Clim.* **23**: 1495–1521, doi: 10.1002/joc.945.

Moberg A, Tuomenvirta H, Nordli Ø. 2005. Recent climatic trends. In *The Physical Geography of Fennoscandia*. Matti Seppälä, Oxford University Press: Oxford, UK, 113–133.

Mohn H. 1867. *Meteorologisk Aarbog (Year book of the Norwegian Meteorological Institute)*. Norwegian Meteorological Institute: Oslo.

Mohn H. 1872. *Meteorologisk Aarbog (Year book of The Norwegian Meteorological Institute)*. Norwegian Meteorological Institute: Oslo.

Mohn H. 1876. *Jahrbuck des Norwegischen Meteorologischen Institut (Year book of the Norwegian Meteorological Institute)*. Norwegian Meteorological Institute: Oslo.

Mohn H. 1895. *Klimatabeller for Norge. I. Luftens temperatur (Climate Tables for Norway. I. Air Temperature)*. Vidensk. Skr. I. Mathem.-naturv. Klasse. Videnskabsselskabets Skrifter: Oslo.

Mohn H. 1910. *Jahrbuck des Norwegischen Meteorologischen Institut für 1910 (Year book of the Norwegian Meteorological Institute)*. Norwegian Meteorological Institute: Oslo.

Nordli Ø. 1997. Homogenitetstesting av norske temperaturseriar II (Homogeneity testing of Norwegian temperature series). DNMI-rapport No. 29/97Klima, MET Norway, Oslo.

Nordli PØ. 2001. Reconstruction of nineteenth century summer temperatures in Norway by proxy data from farmers' diaries. *Clim. Change* **48**: 201–218, doi: 10.1023/A:1005698302572.

Nordli Ø, Alexandersson H, Frich P, Førland EJ, Heino R, Jónsson T, Tuomenvirta H, Tveito OE. 1997. The effect of radiation screens on Nordic time series of mean temperature. *Int. J. Clim.* **17**: 1667–1681, doi: 10.1002/(SICI)1097-0088(199712)17:15<1667::AID-JOC221>3.0.CO;2-D.

Nordli Ø, Tveito OE, 2008. Calculation of monthly mean temperature by Köppen's formula in the Norwegian station network. Met-No Report, MET Norway, Oslo, 13 pp.

Norw. Met. Inst. 1887. Meteorologiske Iagtagelser paa Christiania Observatorium 1887 (Meteorological Observations at Christiania Observatory 1887). Handwritten protocol, MET Norway, Oslo.

© 2014 Royal Meteorological Society

10970088, 2015, 12, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4223 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Norw. Met. Inst. 1920. *Jahrbuch des Norwegischen Meteorologischen Instituts (Year book of the Norwegian Meteorological Institute)*. Norwegian Meteorological Institute: Oslo.

Parker DE. 1994. Effects of changing exposure of thermometers at land stations. *Int. J. Climatol*. **14**, 1–31, doi: 10.1002/joc.3370140102.

Parker D, Horton B. 2005. Uncertainties in central England temperature 1878–2003 and some improvements to the maximum and minimum series. *Int. J. Clim.* **25**: 1173–1188, doi: 10.1002/joc.1190.

R Development Core Team 2004. R: A language and Environment for Statistical Computing, Vienna, Austria. http://www.R-project.org (accessed 24 November 2014).

Torrence C, Compo GP. 1998. A practical guide to wavelet analysis. *Bull. Am. Meteorol. Soc.* **79**: 61–78, doi: 10.1175/1520-0477(1998)079<0061:APGTWA>2.0.CO;2.

Tuomenvirta H. 2001. Homogeneity adjustments of temperature and precipitation series – Finnish and Nordic data. *Int. J. Clim.* **21**: 495–506, doi: 10.1002/joc.616.

van Engelen AFV, Geurts HAM. 1985. *Nicolaus Cruquius (1678-1754) and his meteorological observations (No. K.N.M.I., 165-IV)*. Koninklijk Nederlands Meteorologisch Instituut: De Bilt, Netherlands.

Venema V, Mestre O, COST-Action 0601 Author Team. 2011. Description of the COST-HOME monthly benchmark dataset and the submitted homogenized contributions (COST ACTION-ES0601). University of Bonn: Bonn, Germany

Vincent LA. 1998. A technique for the identification of inhomogeneities in canadian temperature series. *J. Clim.* **11**: 1094–1104, doi: 10.1175/1520-0442(1998)011<1094:ATFTIO>2.0.CO;2.

Wilks DS. 1995. *Statistical Methods in the Atmospheric Science*. *An Introduction, International Geophysics Series*. Academic Press: London.

Winkler P. 2009. Revision and necessary correction of the long-term temperature series of Hohenpeissenberg, 1781–2006. *Theor. Appl. Clim.* **98**: 259–268, doi: 10.1007/s00704-009-0108-y.

© 2014 Royal Meteorological Society

10970088, 2015, 12, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4223 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License